



Lakshya JEE 2.0 (2024)

Matrices

DPP-02

- If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$, then:
 - $\alpha = 2ab, \beta = a^2 + b^2$
 - $\alpha = a^2 + b^2, \beta = ab$
 - $\alpha = a^2 + b^2, \beta = 2ab$
 - $\alpha = a^2 + b^2, \beta = a^2 - b^2$
- If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ a & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where I is 3×3 identity matrix, then the ordered pair (a, b) is equal to:
 - $(2, 1)$
 - $(-2, -1)$
 - $(2, -1)$
 - $(-2, 1)$
- The total number of matrices $A = \begin{pmatrix} 0 & 2y & 1 \\ 2x & y & -1 \\ 2x & -y & 1 \end{pmatrix}$ ($x, y \in \mathbb{R}, x \neq y$) for which $A^T A = 3I_3$ is:
 - 3
 - 4
 - 2
 - 6
- If $A = \begin{bmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{bmatrix}$, then $A + A' = I$, if value of α is
 - $\frac{\pi}{6}$
 - $\frac{\pi}{3}$
 - π
 - $\frac{3\pi}{2}$
- If $A = \text{diag}(1, -1, 2)$ and $B = \text{diag}(2, 3, -1)$, then find $3A + 4B$.
 - Diag (11, 8, 2)
 - Diag (11, 9, 2)
 - diag (11, 2, 9)
 - diag (11, 10, 2)
- Let $A + 2B = \begin{bmatrix} 1 & 2 & 0 \\ 6 & -3 & 3 \\ -5 & 3 & 1 \end{bmatrix}$ and $2A - B = \begin{bmatrix} 2 & -1 & 5 \\ 2 & -1 & 6 \\ 0 & 1 & 2 \end{bmatrix}$ then find $\text{Tr}(A) - \text{Tr}(B)$.
 - 1
 - 2
 - 3
 - 4
- Which of the following is incorrect
 - $\text{Tr}(kA) = k \text{Tr}(A)$ where k is scalar
 - $\text{Tr}(A + B) = \text{Tr}(A) + \text{Tr}(B)$
 - $\text{Tr}(A - B) = \text{Tr}(A) - \text{Tr}(B)$
 - $\text{Tr}(A) = n$, where A is scalar matrix
- The value of x for which the matrix product $\begin{bmatrix} 2 & 0 & 7 \\ 0 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix} \begin{bmatrix} -x & 14x & 7x \\ 0 & 1 & 0 \\ x & -4x & -2x \end{bmatrix}$ equals an identity matrix, is
 - $1/2$
 - $1/3$
 - $1/4$
 - $1/5$
- If $A = \begin{bmatrix} -4 & -1 \\ 3 & 1 \end{bmatrix}$, then the determinant of the matrix $(A^{2016} - 2A^{2015} - A^{2014})$, is
 - 2014
 - 2016
 - 175
 - 25
- If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and $A \text{ adj } A = AA^T$, then $5a + b$ is equal to:
 - 1
 - 5
 - 4
 - 13



Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

1. (3)
2. (2)
3. (2)
4. (2)
5. (2)

6. (2)
7. (4)
8. (4)
9. (4)
10. (2)



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>